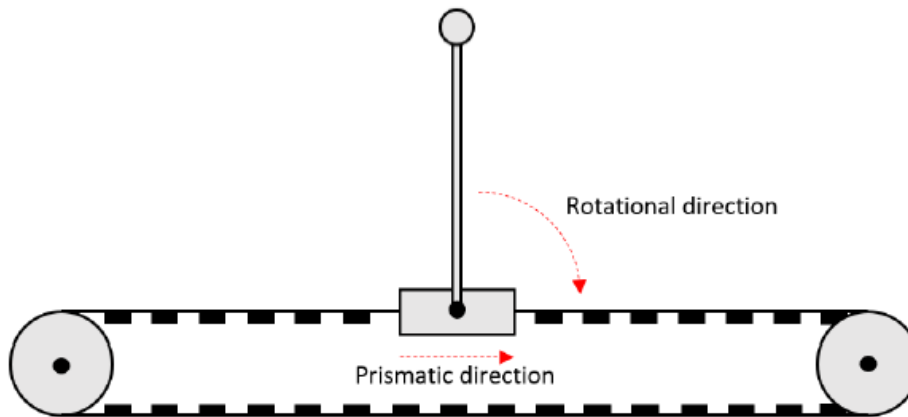


System parameters



- The mass of the cart = 0.5 [kg]
- The mass of the rod = 0.082 [kg]
- The mass of the weight attached to the pendulum = 0.002 [kg]
- The mass of the pendulum = 0.084 [kg]
- The total length of the rod = 0.35 [m]
- The gravitational acceleration is assumed to be $g = 9.82$ [m/s²]
- Radii of the belt rollers = 0.05 [m]
- Length of the conveyor belt = 1.72 [m]
- Linear damping coefficient of the conveyor belt = 5 [N/(m/s)]
- Rotational damping coefficient of the pendulum = 0.0012 [Nm/(rad/s)] (i.e., the joint between the cart and pendulum)